

standard_module_outline_design

Module Outline Design Standard

Purpose

Defines the required structure for any curriculum module outline produced for Junior Backend Developer training, whether authored by a human curriculum designer or generated by an automated system. This standard exists to ensure that every module — regardless of who or what produced it — can be evaluated consistently against the same structural quality bar, and can be traced back to the specific competency gap it was designed to address.

Required Sections

Every module outline must include the following sections, in the order listed below, so that reviewers evaluating multiple module outlines can scan and compare them consistently:

1. **Target Competency** — the specific competency, referencing the Backend Competency Framework by its numbered identifier (for example, “2.1 Relational Database Design”), that this module is designed to address. A module may target more than one competency only if it explicitly justifies why the competencies are naturally taught together, rather than combining unrelated competencies for convenience.
2. **Prerequisite Knowledge** — what a learner must already know before starting this module, stated specifically enough to be checkable (for example, “comfortable writing basic SELECT and JOIN queries” rather than simply “basic SQL knowledge”).
3. **Learning Objectives** — specific, measurable statements of what the learner will be able to do after completing the module, phrased in terms of observable behavior (for example, “design a normalized schema for a given set of business requirements and justify each normalization decision”) rather than vague internal states (such as “understand database design”).
4. **Content Outline** — a topic-by-topic or session-by-session breakdown of material covered, sufficiently detailed that a different curriculum designer could estimate the module’s total duration and identify any obvious gaps between the content outline and the stated learning objectives.
5. **Assessment Alignment** — an explicit statement of how the module’s content will be assessed, and specifically which assessment items or item types map to which learning objectives, so that every stated objective has at least one corresponding point of assessment.

Quality Checks

A module outline should be rejected or sent back for revision if any of the following apply:

- Learning objectives are vague or unmeasurable (for example, “understand APIs” rather than “design a resource-oriented API endpoint for a described feature, applying correct HTTP methods and status codes”).
- The content outline does not clearly and traceably lead to the stated learning objectives — for example, a learning objective concerning query optimization with no corresponding content

- section addressing query plans or indexing.
- No assessment is proposed for one or more stated learning objectives, or the proposed assessment does not actually map back to what the objective claims to measure.
- The target competency reference is missing, incorrect, or refers to a competency that does not exist in the current version of the Backend Competency Framework.

Gap-to-Module Traceability

When a module is generated in direct response to an identified competency gap (as opposed to being part of a general foundational curriculum), the outline must explicitly reference which specific gap it addresses, including the gap's recorded evidence where available. This traceability requirement allows the overall curriculum plan for a given individual or cohort to be audited for completeness: every identified gap should map to at least one module addressing it, and every module in a gap-driven curriculum plan should map back to at least one identified gap, with no orphaned modules that do not correspond to any actual identified need.

Handling Ambiguous or Overly Broad Gaps

If a competency gap analysis identifies a gap that is too broad to address with a single module (for example, a gap spanning an entire competency domain rather than a specific sub-competency), the module designer should decompose the gap into multiple, narrower modules, each separately traceable to a specific sub-competency, rather than producing a single overly broad module that attempts to cover too much ground to be assessed meaningfully.

Owner

Curriculum Design Standards Committee, in coordination with whichever team or system (human or automated) is responsible for generating module outlines in a given curriculum design workflow.